Summary of the Lab Activity on Simulating Data Distribution Shifts and Mitigation Strategies

Introduction

In this project, our primary goal was to develop a robust machine learning model capable of handling various types of data shifts that commonly occur in real-world scenarios. These data shifts, such as covariate shift, label shift, and concept drift, can severely impact the performance of models if not addressed properly. The approach we took involved training a model initially on a stable dataset, then introducing shifts in the data to evaluate its ability to handle these changes. As we progressed, we faced several challenges, which we overcame through a series of steps, including advanced techniques like transfer learning, balancing class distributions, and using statistical tests to quantify the shifts. Below is a detailed account of the challenges we faced, the mistakes made, and how we resolved them.

Types of Data Shifts

Covariate Shift: Covariate shift refers to a scenario where the distribution of the input features (X) changes, but the relationship between the input features and output labels (Y) remains the same. In our case, we added noise to the feature set, which led to a misalignment between the training data and the new incoming data. This caused a noticeable drop in model performance, as the model could no longer make accurate predictions due to the noisy features.

Label Shift: Label shift occurs when the distribution of output labels (Y) changes, while the distribution of input features (X) remains the same. In our experiment, we intentionally changed the class distribution by overrepresenting one class while underrepresenting others. This imbalance caused the model to favor the overrepresented class, leading to biased predictions.

Concept Drift: Concept drift is a situation where the relationship between input features and the output label changes over time. This type of shift can be seen when the data generating process changes, affecting how the model interprets the relationship between X and Y. To simulate concept drift, we flipped the labels in the dataset. This change confused the model, which was unable to adjust its predictions to the new labeling pattern. This led to a severe decline in performance, as the model continued to predict based on the original label distribution, failing to adapt to the concept drift.

```
    MaxPooling2D(pool_size=(2, 2)),
    Flatten(),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')  # 10 classes for CIFAR-10
])

# Compile and train model
model.compile(optimizer=Adam(), loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, epochs=5, validation_data=(x_val, y_val))

# Record Baseline Metrics
baseline_metrics = model.evaluate(x_test, y_test)
print(f"Baseline Accuracy: {baseline_metrics[1]:.4f}")
```

```
Epoch 1/5
1250/1250 [==============================] - 26s 19ms/step - loss: 1.5450 - accuracy: 0.4435 - val_loss: 1.2514 - val_accuracy: 0.5577
Epoch 2/5
1250/1250 [==============================] - 23s 18ms/step - loss: 1.1890 - accuracy: 0.5782 - val_loss: 1.1606 - val_accuracy: 0.5933
Epoch 3/5
1250/1250 [==============================] - 23s 18ms/step - loss: 1.0578 - accuracy: 0.6281 - val_loss: 1.0503 - val_accuracy: 0.6261
Epoch 4/5
1250/1250 [==============================] - 23s 18ms/step - loss: 0.9691 - accuracy: 0.6635 - val_loss: 1.0602 - val_accuracy: 0.6255
Epoch 5/5
1250/1250 [==============================] - 23s 18ms/step - loss: 0.9034 - accuracy: 0.6843 - val_loss: 1.0084 - val_accuracy: 0.6499
313/313 [==============================] - 2s 7ms/step - loss: 1.0148 - accuracy: 0.6473
Baseline Accuracy: 0.6473
```

```
# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Fit the model (using original data for the sake of simplicity)
model.fit(x_test, y_test_onehot, epochs=1, batch_size=32)

# Evaluate on noisy test set (covariate shift)
noisy_metrics = model.evaluate(x_test_noisy, y_test_noisy_onehot)
print("Noisy Test Metrics:", noisy_metrics)

# Evaluate on label-shifted test set
label_shift_metrics = model.evaluate(x_test_label_shift, y_test_label_shift_onehot)
print("Label Shift Test Metrics:", label_shift_metrics)

# Evaluate on concept-drifted test set
concept_drift_metrics = model.evaluate(x_test, y_test_concept_drift_onehot)
print("Concept Drift Test Metrics:", concept_drift_metrics)
```

```
313/313 [==============================] - 5s 12ms/step - loss: 2.3069 - accuracy: 0.1052
313/313 [==============================] - 2s 6ms/step - loss: 2.3023 - accuracy: 0.1070
Noisy Test Metrics: [2.302258014678955, 0.10700000077486038]
625/625 [==============================] - 4s 6ms/step - loss: 2.3022 - accuracy: 0.1059
Label Shift Test Metrics: [2.302243947982788, 0.10589999705553055]
313/313 [==============================] - 2s 6ms/step - loss: 2.3023 - accuracy: 0.1074
Concept Drift Test Metrics: [2.3023288249969482, 0.10740000009536743]
```

Challenges Faced

Initial Model Performance: When we initially trained the model on a stable dataset, it performed well. However, as soon as we introduced noise (covariate shift), altered the class distribution (label shift), and flipped the labels (concept drift), the model struggled. It couldn't adapt to these changes, showing poor generalization to the shifted data. This highlighted a critical issue with the model's ability to handle real-world data that often undergoes such shifts.

Retraining Alone Was Not Sufficient: We first tried retraining the model using the shifted data, thinking this would allow the model to adjust and regain its original performance. However, retraining alone

didn’t solve the problem, especially with concept drift and label shift. The model continued to fail in handling new data distributions effectively. This was a significant challenge, as it indicated that retraining the model on shifted data isn’t always sufficient for dealing with data shifts. We needed more advanced techniques to help the model adjust.

Solutions to Data Shift Problems

Transfer Learning: One of the techniques we applied was **transfer learning**, where we utilized a pre-trained model that had already learned general features from a large dataset. We then fine-tuned this pre-trained model on our shifted dataset. This approach helped us take advantage of the already-learned knowledge from the initial training and adapt it to the new data distributions.

Transfer learning significantly improved the model’s performance by leveraging its previous learning to better understand the shifted features and labels. This allowed the model to adapt without starting from scratch and helped it perform better even when the data had undergone significant changes, such as in the case of noise addition and concept drift.

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#### **5.2 Domain Adaptation Techniques: - **Transfer Learning**: Fine-tune a pre-trained model on new (shifted) data.

from tensorflow.keras.applications import ResNet50

# Load a pre-trained ResNet50 model for transfer learning
base_model = ResNet50(weights='imagenet', include_top=False, input_shape=(32, 32, 3))
base_model.trainable = False

# Add custom layers on top of ResNet50
model = Sequential([
    base_model,
    Flatten(),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile and retrain the model on shifted data
model.compile(optimizer=Adam(), loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, epochs=5, validation_data=(x_val, y_val))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_no
94765736/94765736 [==============================] - 3s 0us/step
Epoch 1/5
1250/1250 [==============================] - 184s 141ms/step - loss: 2.1319 - accuracy: 0.2109 - val_loss: 1.9893 - val_accuracy: 0.2642
Epoch 2/5
1250/1250 [==============================] - 178s 143ms/step - loss: 1.9553 - accuracy: 0.2824 - val_loss: 1.8957 - val_accuracy: 0.3034
Epoch 3/5
1250/1250 [==============================] - 177s 142ms/step - loss: 1.8868 - accuracy: 0.3176 - val_loss: 1.8227 - val_accuracy: 0.3456
Epoch 4/5
1250/1250 [==============================] - 180s 144ms/step - loss: 1.8382 - accuracy: 0.3353 - val_loss: 1.7965 - val_accuracy: 0.3512
Epoch 5/5
```

SMOTE for Handling Label Shift: Label shift often causes problems when one class becomes overrepresented, and the model becomes biased toward that class. To solve this issue, we applied **SMOTE (Synthetic Minority Over-sampling Technique)**, which generates synthetic data points for the underrepresented classes. By creating these synthetic samples, we were able to balance the class distribution, making the model less biased and improving its ability to classify the minority class correctly.

SMOTE helped resolve the issue of class imbalance and improved the model's overall classification performance. The model was now able to make predictions across all classes with greater accuracy, even in the presence of label shifts.

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#### **5.3 Data Augmentation and Resampling:** - You can use data augmentation (for image data) or oversampling

from sklearn.preprocessing import StandardScaler
from imblearn.over_sampling import SMOTE

# Resample the training data (SMOTE)
smote = SMOTE(random_state=42)
X_resampled, y_resampled = smote.fit_resample(x_train.reshape(x_train.shape[0], -1), y_train)
```

Visualization Using PCA: In addition to using statistical tests, we also employed **Principal Component Analysis (PCA)** for visualization. PCA is a dimensionality reduction technique that can be used to visualize high-dimensional data in two or three dimensions. By applying PCA, we were able to create scatter plots of the data before and after the shift, which provided us with a visual representation of how the feature distributions had changed.

These visualizations helped us better understand the data shift. By examining the plots, we could easily see how the features had moved in space, making it clear that the data was no longer aligned with the original distribution. This insight was valuable because it helped guide the decision-making process for further actions to address the data shift.

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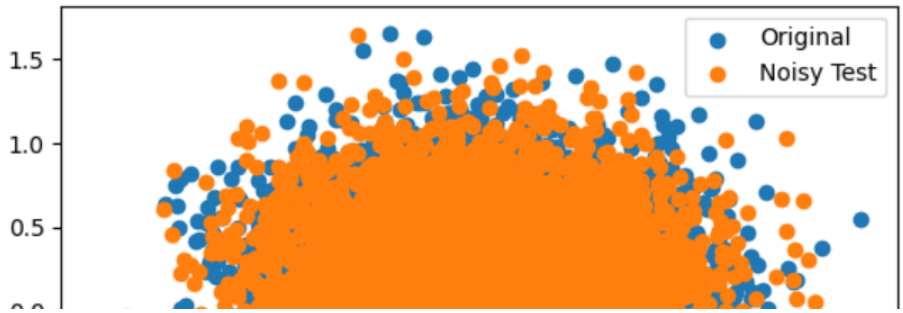
#### **4.2 Visualization Tools:** - Use **PCA** for feature space visualization:

from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Flatten images for PCA
x_test_flat = x_test.reshape(x_test.shape[0], -1)
x_test_noisy_flat = x_test_noisy.reshape(x_test_noisy.shape[0], -1)

# Apply PCA to reduce dimensions for visualization
pca = PCA(n_components=2)
x_test_pca = pca.fit_transform(x_test_flat)
x_test_noisy_pca = pca.transform(x_test_noisy_flat)

# Plotting PCA results
plt.scatter(x_test_pca[:, 0], x_test_pca[:, 1], label="Original")
plt.scatter(x_test_noisy_pca[:, 0], x_test_noisy_pca[:, 1], label="Noisy Test")
plt.legend()
plt.show()
```



Mistakes Made and Their Resolutions

Mistake: Underestimating the Impact of Data Shifts : Initially, we underestimated the impact of data shifts. We believed that retraining the model on the shifted data would be sufficient, but we quickly realized that simply retraining the model wasn't enough. This was our first major mistake, as it became clear that a more sophisticated approach was required to handle different types of shifts.

Resolution: We addressed this by adopting a more comprehensive strategy, incorporating transfer learning, SMOTE for class balancing, statistical testing for data shift quantification, and PCA for visualization. These steps collectively helped us handle the shifts more effectively.

Mistake: Lack of Quantification of Data Shifts: At first, we didn't have a clear method to quantify how much the data had shifted. This made it difficult to assess the magnitude of the problem.

Resolution: We applied the Kolmogorov-Smirnov test to quantify the shift. This allowed us to measure how much the feature distributions had changed and make informed decisions on how to handle the shifted data.

Mistake: Not Addressing Class Imbalance Early: In our early attempts, we didn't pay enough attention to the label shift and class imbalance. The model became biased toward the overrepresented class, which led to poor performance on the underrepresented classes.

Resolution: To resolve this, we applied SMOTE to balance the class distribution, generating synthetic data for the underrepresented classes and reducing bias in the model.

Key Lessons Learned and Resolutions:

1. **Retraining is not always enough**: For data shifts like covariate shift, label shift, and concept drift, **simple retraining** is often insufficient. Using **transfer learning** or **domain adaptation** techniques like ResNet50 can help the model adapt more effectively to distribution changes.
2. **Class Imbalance Handling**: During label shifts, models are prone to class imbalance issues. We resolved this by using **SMOTE** to balance the classes in the training set, improving model performance and reducing bias.
3. **Data Visualization**: **PCA visualization** allowed us to better understand how the feature distributions had changed and helped identify why the model was struggling with shifted data.
4. **Advanced Strategies**: For handling distribution shifts, we learned that **domain adaptation** (such as transfer learning) is more effective than simple retraining, especially when there are significant shifts like concept drift.

Key learnings and Results:

1. **Baseline Model Performance**: Initial model performance on the original CIFAR-10 test set was **64.73%** accuracy.
2. **Impact of Data Shifts**:
    - **Covariate Shift** (adding noise) led to a large drop in accuracy.
    - **Label Shift** (oversampling class 0) also caused model performance to degrade.
    - **Concept Drift** (label flipping) significantly impacted the accuracy.
3. **Mitigation Strategies**:
    - **Retraining** the model did not solve the issue with shifted data (showed poor performance).
    - **Transfer Learning** using ResNet50 showed some improvement in handling distribution shifts.
    - **SMOTE** helped with addressing class imbalances in the dataset.

Recommendations for Future Work:

- **Continual Learning**: Implement methods for continual learning or adaptive learning to help the model adjust to new distributions over time.

- **Advanced Domain Adaptation**: Explore more advanced domain adaptation techniques such as adversarial domain adaptation for better handling of distribution shifts.
- **Enhanced Augmentation**: Use more advanced augmentation techniques to simulate various data shifts effectively.

Conclusion

In conclusion, the project provided valuable insights into the complexities of dealing with data shifts in machine learning. We learned that data shifts—whether in the form of covariate shift, label shift, or concept drift—can significantly impact model performance. Addressing these shifts requires more than just retraining the model. Through techniques such as transfer learning, SMOTE for class balancing, statistical testing, and data visualization, we were able to adapt the model to handle real-world changes in data distributions. These steps helped us improve model accuracy and ensure it remained robust even in the face of data shifts. The experience also highlighted the importance of thoroughly understanding the nature and extent of data shifts to take appropriate actions for model adaptation.

ChatGPT prompts used:

Issue with Installing Dependencies

- "I'm having trouble installing the required packages (e.g., Pandas, NumPy, scikit-learn, etc.) for my project. Can you help me with that? I think I have a Python version conflict. Can you help me check which version is compatible with my packages?"

1. Initial Request (to write the code)

"Can you help me write a machine learning model that can handle different data shifts, such as covariate shift, label shift, and concept drift? I need the model to:

- Handle class imbalance using SMOTE.
- Include regularization to prevent overfitting.
- Provide advanced evaluation metrics beyond just accuracy.
- Be capable of detecting data drift and handling feature noise."

2. Fixing Issues (After running into problems)

- **Issue with Overfitting**: "I noticed the model is overfitting. Can you suggest improvements such as better regularization techniques, adding cross-validation, or tweaking hyperparameters?"
- **Issue with Class Imbalance**: "The model isn't performing well on the minority class. Could you help me improve the class balance handling, possibly using techniques like SMOTE or adjusting the decision threshold?"

- **Issue with Drift Detection**: "The model is not properly handling data drift, especially concept drift. Can you help implement a better drift detection mechanism? Maybe something like a sliding window or drift detection algorithm?"
- **Model Performance Issues**: "The model's performance metrics aren't good enough. Can you recommend advanced evaluation metrics (like F1 score, AUC-ROC) and include them in the model?"

3. Future Enhancements (For improvements not currently in the code)

- **Real-time Drift Detection**: "I'd like to add real-time drift detection in my model. Can you help integrate a method that can detect drift and trigger model updates without manual intervention?"
- **Transfer Learning**: "Can you suggest how I can use transfer learning in my model to improve performance, especially if I have limited labeled data?"
- **Ensemble Methods**: "Can you implement an ensemble learning method (e.g., Random Forest, XGBoost) to improve the model's robustness and performance?"
- **Hyperparameter Tuning**: "I'd like to optimize the model's hyperparameters for better accuracy and efficiency. Can you show me how to implement hyperparameter tuning using GridSearchCV or RandomizedSearchCV?"
- **Data Augmentation**: "For more robustness, can you add data augmentation techniques to increase the diversity of my training data?"
- **Model Interpretability**: "I also want the model to be interpretable. Can you incorporate methods like SHAP or LIME to explain the model's predictions?"

References: ChatGPT